

Technology Stack

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit (Python)	Provides a simple, interactive, and responsive web interface for audio upload, playback, visualization, and displaying analysis results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.10+	Handles speech processing, semantic analysis, audio feature extraction, scoring, and report generation with strong AI and machine learning library support.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite	Stores uploaded audio metadata, transcripts, evaluation scores, session details, and generated reports. It is lightweight and easy to integrate.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Streamlit Community Cloud <i>(or Local Deployment)</i>	Enables easy deployment and sharing of the application with minimal configuration. Suitable for educational and prototype projects.

5	Version Control & CI/CD	Git & GitHub	Supports version control, team collaboration, code management, and project documentation throughout development.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	OpenAI Whisper, Sentence-BERT, Librosa, Google Gemini API, ReportLab, Matplotlib, SoundFile, NLTK	Whisper performs speech-to-text transcription. Sentence-BERT evaluates semantic similarity. Librosa extracts audio features. Gemini generates intelligent feedback. ReportLab creates PDF reports. Matplotlib visualizes waveforms. SoundFile handles audio files. NLTK supports text preprocessing and language analysis.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		